EEI3346 Mini Project Report

Name :K.L.S.K.Dilshan
Number : s24016290

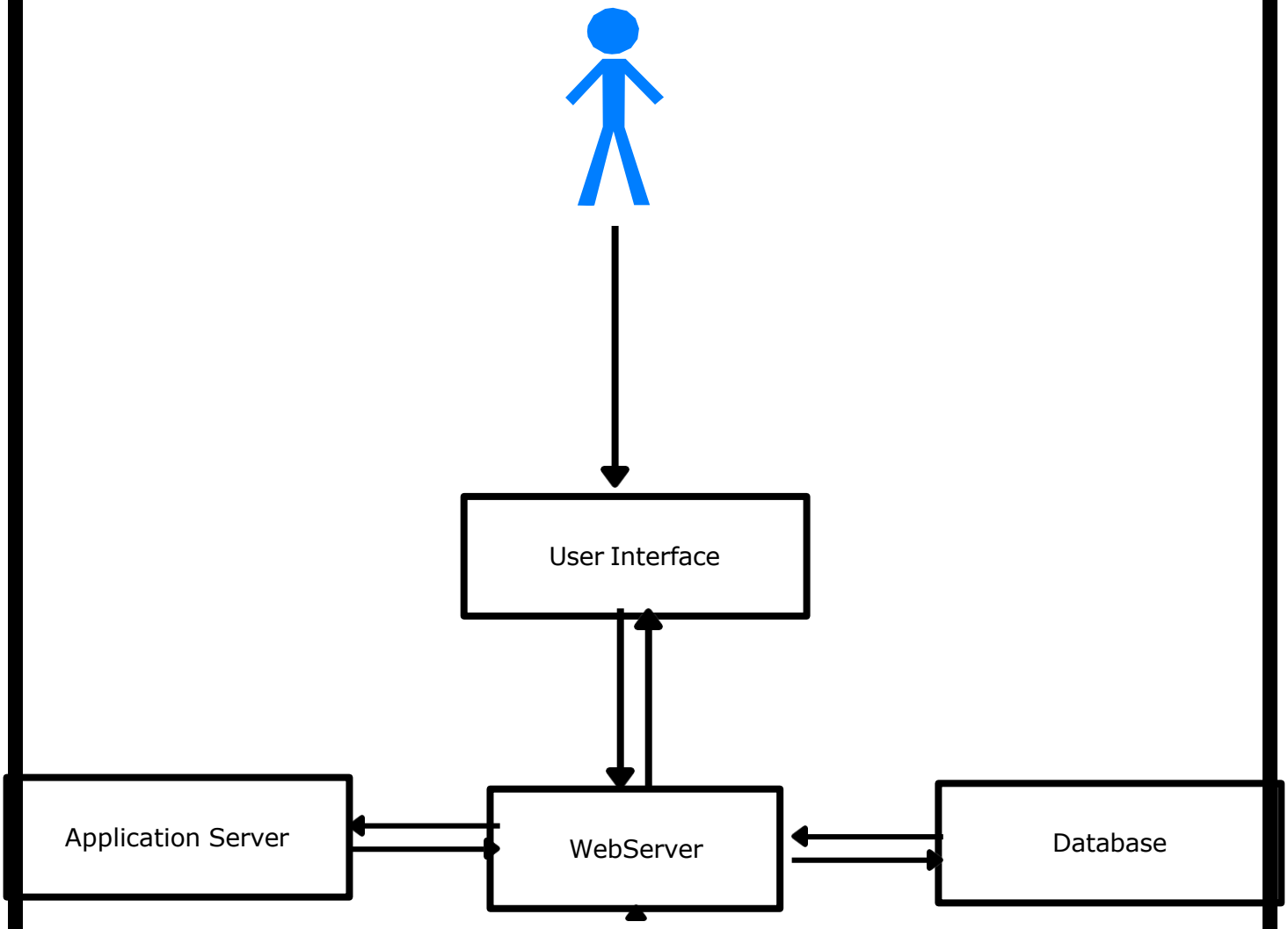
Problems

- 1 problem I faced is using Vue Js to create feedback section in my web page(Page was Strucked).
- 2 problem I faced : I want to convert navigation bar to drop down menu in mobile view mode.
- 3 problem I faced Includs CSS Wrong in code.

Solution Approach

- I learn vue js from youtube videos and some other educational sites and then I created it as I can.
- To solve second problem I watched youtube video and learn How to create drop down menu.Then I solve my problem perfectly.
- Some Times Use Internet And Asked My friends.

Architectural diagram of the system



Site Map

Used web technologies and examples

- **HTML**

Used to create the website's main sketch.

- **CSS**

Used on all webpages to give them a polished appearance.

- **Java Script**

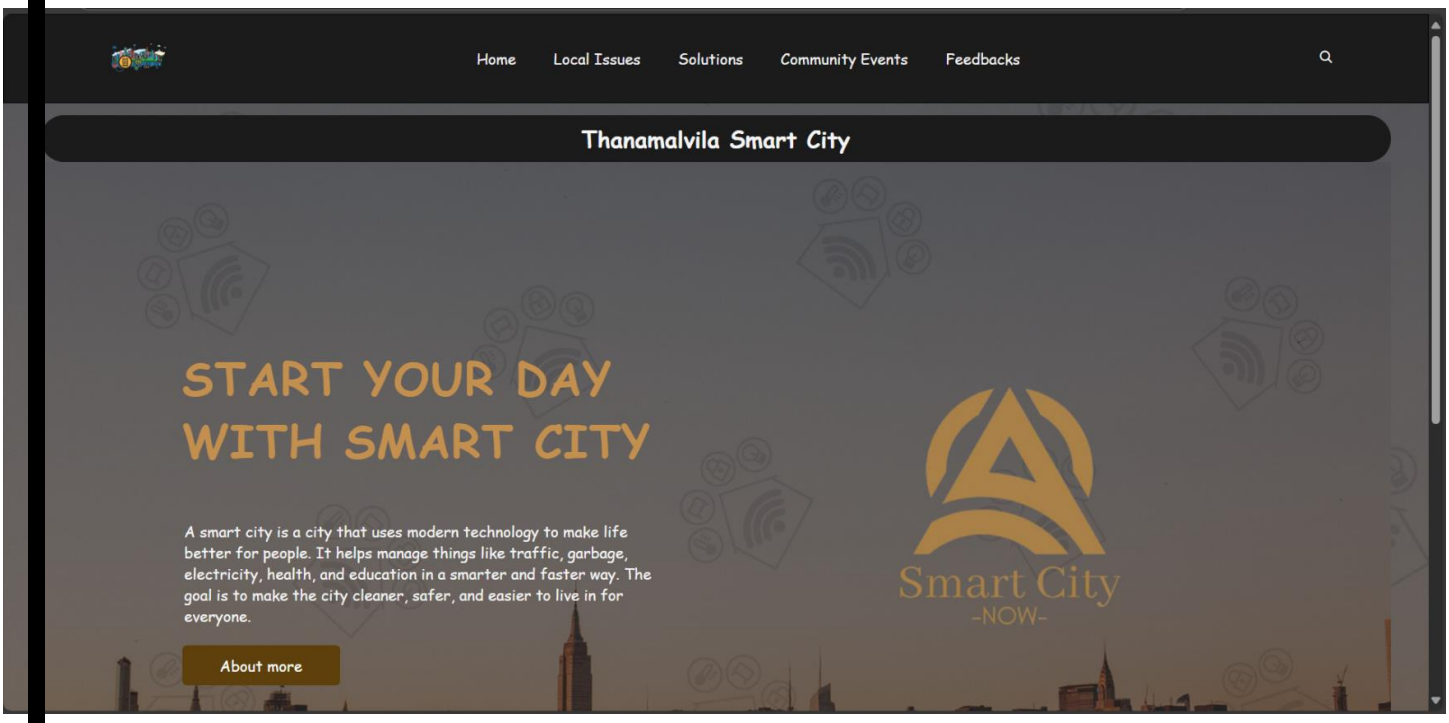
Use to create a drop-down menu in mobile mode.

- **Vue JS**

Use for create Feedbacks section in the web site

Test Results

Home



Local Issues

Home Local Issues Solutions Community Events Feedbacks

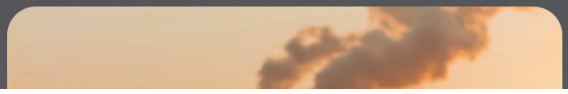
LOCAL ISSUES

Garbage Problem

Uncollected Waste in Streets --Garbage is often left on roads and in public places without being collected. This makes the city look dirty and can block drains during rain.

Lack of Bins and Recycling --There aren't enough dustbins in many areas, so people throw waste on the ground. Recycling is also not common, which increases pollution.

Health and Environment Issues --Open garbage attracts flies, mosquitoes, and stray animals. It causes bad smells, spreads diseases, and harms the environment.



Health Problems

Air Pollution --Smoke from vehicles, factories, and burning

Home Local Issues Solutions Community Events Feedbacks

Health Problems

Air Pollution --Smoke from vehicles, factories, and burning garbage makes the air dirty. Breathing this polluted air can cause coughing, asthma, and other lung problems.

Poor Sanitation --Some areas don't have clean toilets or proper drainage systems. This can lead to the spread of diseases like dengue, diarrhea, and skin infections.

Lack of Public Health Awareness --Many people don't know how to stay healthy or prevent illness. Without health education, basic habits like washing hands or using clean water are often ignored.

Education Issues

Limited access to digital learning --Many students do not have internet or devices at home. This makes it hard for them to join online classes or use digital learning tools.

Outdated Classrooms --Some schools still use old teaching methods and lack modern equipment. Broken furniture and poor classroom conditions affect how well students learn.

Unequal Quality of Education --Students in rural or low-income areas often get a lower quality of education. They have fewer resources, less experienced teachers, and limited support.

Home Local Issues Solutions Community Events Feedbacks

have fewer resources, less experienced teachers, and limited support.

Subsurface Facility Problems

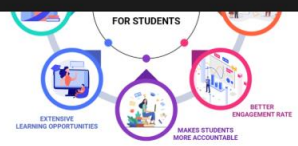
Outdated Water Pipelines and Drainage --Many cities still use old underground water and drainage systems. These pipelines often leak or break, wasting water and causing floods during rain.

Poor Underground Mapping --There is little information about where pipes and cables are buried. Because of this, workers often damage them while digging, leading to more problems.

Frequent Road Repairs --Due to weak underground planning, roads are dug up often to fix hidden issues. This causes traffic, dust, and delays in daily life for residents.

Solutions

Home Local Issues Solutions Community Events Feedbacks



tablets and training teachers to use digital tools also ensures that no child is left behind.By improving digital education, cities can offer equal learning opportunities, support lifelong learning, and build a smarter, more skilled generation.

1-Digital classroom upgrades.
2-Free local Wi-Fi zones.
3-E-learning kiosks in underprivileged areas.

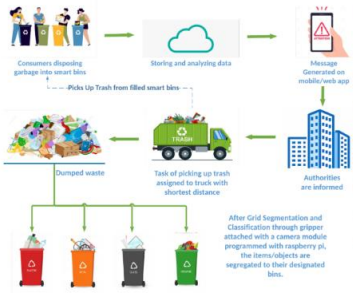
4. Upgrade Subsurface Facilities

Upgrading subsurface facilities involves modernizing underground infrastructure—such as oil wells, mining shafts, or geothermal systems—with advanced technology to boost efficiency, safety, and sustainability. By integrating smart sensors, automated controls, and real-time monitoring, operators can detect leaks, optimize production, and prevent equipment failures before they occur. Data analytics and AI help predict reservoir performance, allowing for better resource management, while robotics and drones enable safer inspections in hard-to-reach areas. These upgrades not only extend the lifespan of subsurface assets but also reduce environmental risks and operational costs, ensuring more reliable and eco-friendly energy and resource extraction.

1-Digital dashboards for utilities.
2-Use GIS (Geographic Information Systems) for mapping underground infrastructure.

Home Local Issues Solutions Community Events Feedbacks

Smart City Solutions



1. Smart Waste Management

Smart Waste Management uses technology to improve the collection, sorting, and recycling of garbage in urban areas. Instead of relying only on fixed schedules, smart bins equipped with sensors can detect when they are full and notify city workers in real time. This prevents overflowing bins and reduces unnecessary collection trips, saving fuel and time.In addition, cities can use GPS tracking for waste trucks to plan efficient routes and avoid traffic, ensuring quicker garbage disposal. Data analytics can also help monitor which areas generate more waste and need better recycling programs. With public awareness campaigns, citizens can also learn how to reduce, reuse, and recycle effectively.Smart waste systems create a cleaner, healthier environment while lowering costs and improving sustainability in modern cities.

1-Smart bins with level sensors.
2-Waste segregation apps.
3-Awareness campaigns.

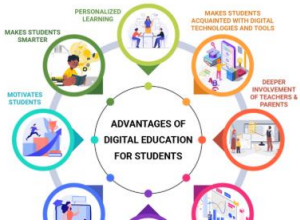
2. Clean Air and Better Sanitation

Clean air and proper sanitation are essential for a healthy city. To reduce air pollution, smart cities promote the use of public transportation, electric vehicles, and green spaces like parks and trees that naturally filter the air. Limiting industrial emissions and encouraging cycling or walking also help keep the air clean and fresh.Better sanitation means building clean public toilets, proper

2. Clean Air and Better Sanitation

Clean air and proper sanitation are essential for a healthy city. To reduce air pollution, smart cities promote the use of public transportation, electric vehicles, and green spaces like parks and trees that naturally filter the air. Limiting industrial emissions and encouraging cycling or walking also help keep the air clean and fresh.Better sanitation means building clean public toilets, proper sewage systems, and efficient drainage to prevent waterlogging and the spread of diseases. Smart sensors can monitor sanitation facilities and alert city workers when maintenance is needed.Together, clean air and improved sanitation make urban areas healthier, more livable, and safer for everyone, especially children and the elderly.

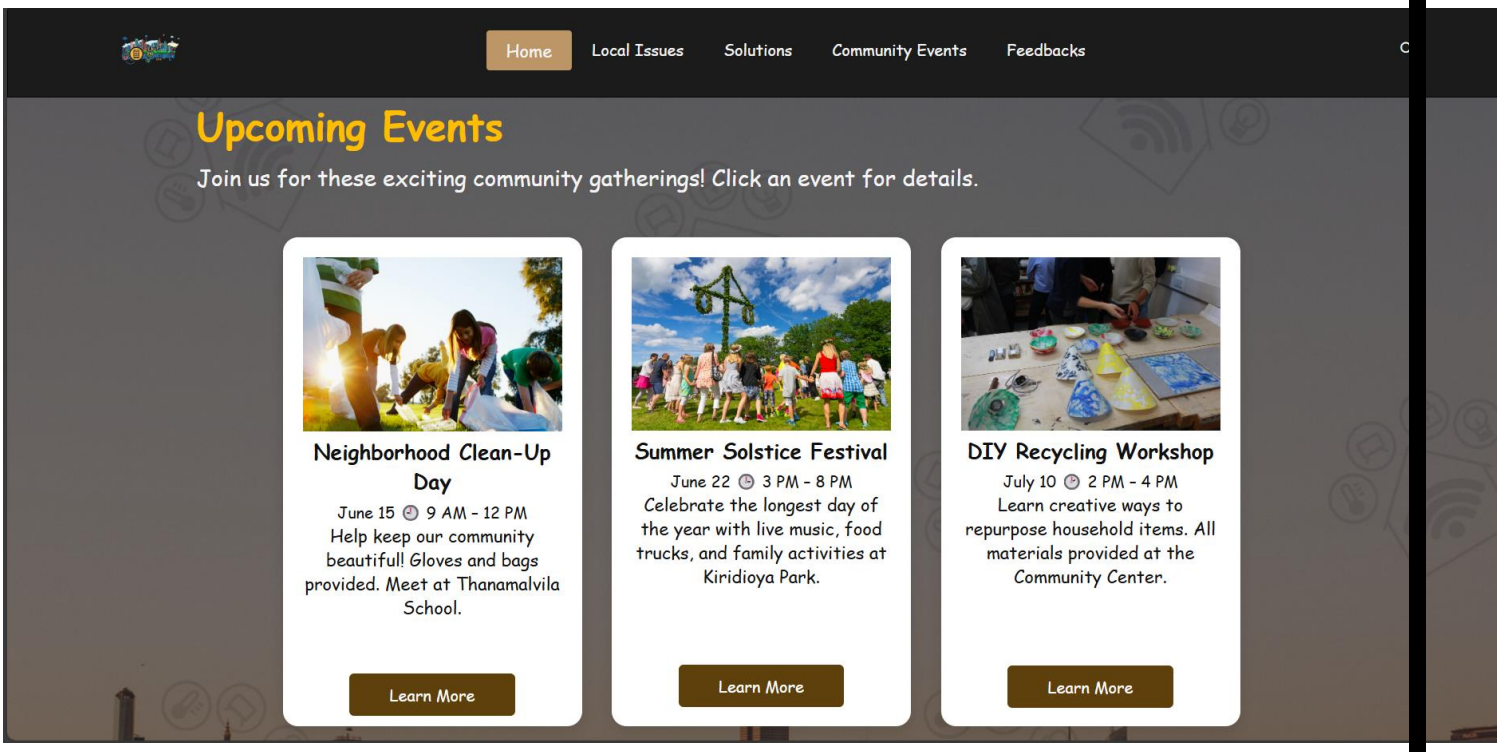
1-Real-time air quality monitors.
2-Public health mobile alerts.
3-Telemedicine units in communities.



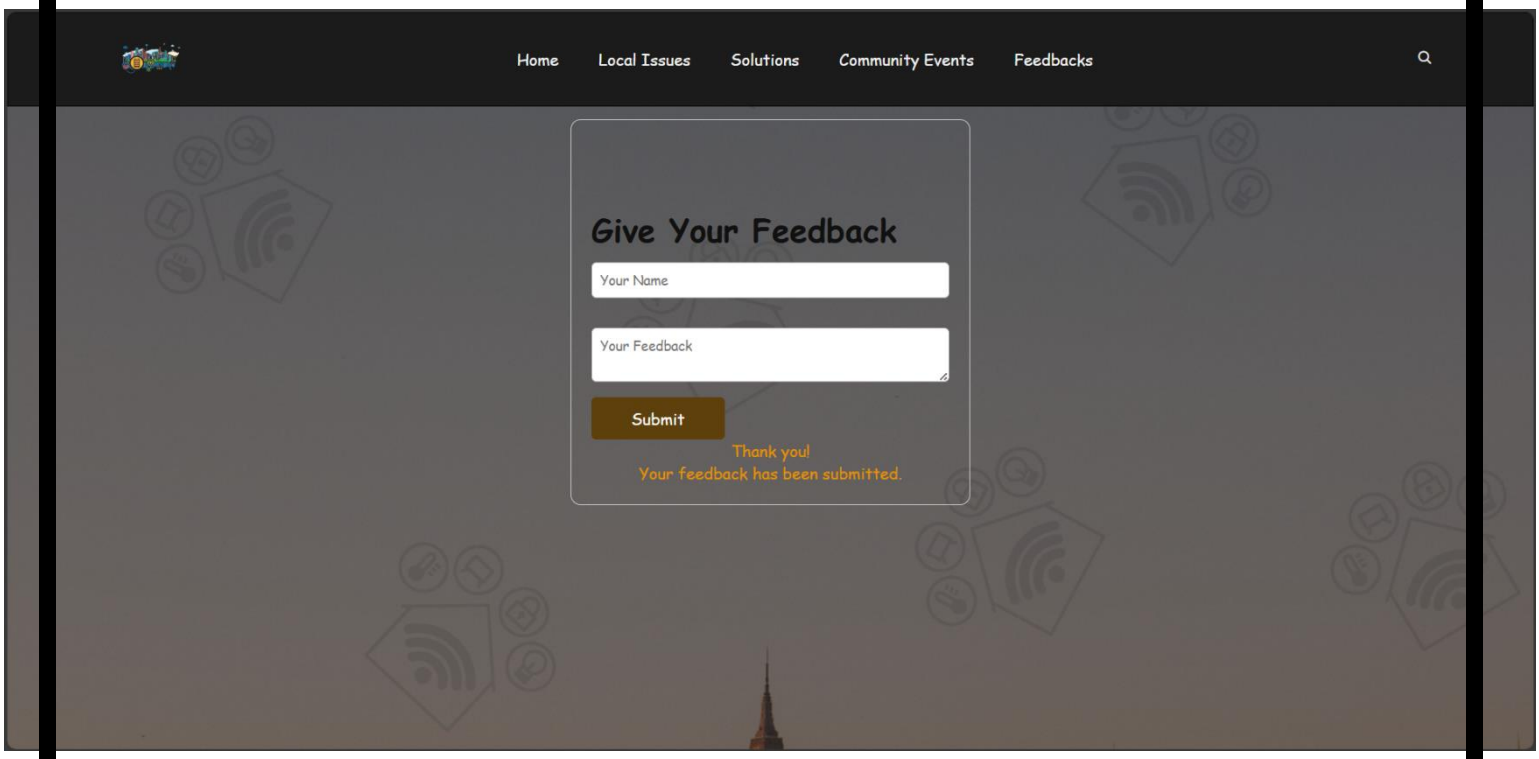
3. Improve Digital Education

Digital education is key to preparing students for the future. In smart cities, schools are equipped with modern tools like smart boards, computers, and internet access to make learning more interactive and effective. These technologies help students understand subjects better and encourage creative thinking.To make sure everyone benefits, online learning platforms and free digital resources should be available for students in rural or low-income areas. Providing devices like tablets and training teachers to use digital tools also ensures that no child is left behind.By improving digital education, cities can offer equal learning opportunities, support lifelong learning, and build a smarter, more skilled generation.

Community Events



Feedbacks



Conclusion

I overcame a number of obstacles to successfully design and develop a working web application for this EEI3346 Web Application Development Mini Project. This project gave me practical experience in debugging, problem-solving, and front-end development.